Supervised Learning: Regression Models and Performance Metrics | Solution

Question 1 : What is Simple Linear Regression (SLR)? Explain its purpose.

Simple Linear Regression (SLR) is a statistical technique used to model the relationship between one independent variable (X) and one dependent variable (Y). It predicts the value of Y based on X using a straight-line equation.

Purpose:
• To understand the relationship between variables.
• To predict future outcomes.
• To identify trends in data.
• To measure how changes in one variable affect another.

Question 2: What are the key assumptions of Simple Linear Regression?

The key assumptions of Simple Linear Regression are:
1. Linearity – There should be a linear relationship between X and Y.
2. Independence – Observations should be independent.
3. Homoscedasticity – Constant variance of residuals.
4. Normality – Residuals should be normally distributed.
5. No significant outliers – Extreme values should not strongly affect the model.

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

The mathematical equation of Simple Linear Regression is:

$Y = a + bX$

Where:
• Y = Dependent variable
• X = Independent variable
• a = Intercept (value of Y when X = 0)
• b = Slope (change in Y for every unit increase in X)
• bX = Contribution of independent variable to prediction

Question 4: Provide a real-world example where simple linear regression can be applied.

A real-world example of Simple Linear Regression is predicting house prices based on house area.

Example:

• Independent Variable (X): Area of house in square feet
• Dependent Variable (Y): Price of house

As the area increases, the price generally increases. A regression model can help estimate house prices for new properties.

Question 5: What is the method of least squares in linear regression?

The method of least squares is a mathematical approach used to find the best-fit regression line by minimizing the sum of squared differences between actual and predicted values.

It helps:
• Reduce prediction error
• Find optimal slope and intercept
• Improve model accuracy

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Logistic Regression is a supervised learning algorithm used for classification problems where the output is categorical, such as Yes/No or 0/1.

Difference between Logistic and Linear Regression:

Linear Regression:
• Used for continuous values
• Output is numerical
• Uses a straight-line equation

Logistic Regression:
• Used for classification
• Output is probability between 0 and 1
• Uses sigmoid function

Question 7: Name and briefly describe three common evaluation metrics for regression models.

1. Mean Absolute Error (MAE):
Measures the average absolute difference between actual and predicted values.

2. Mean Squared Error (MSE):
Measures the average squared difference between actual and predicted values.

3. Root Mean Squared Error (RMSE):
Square root of MSE. It gives error in the same unit as the target variable.

Question 8: What is the purpose of the R-squared metric in regression analysis?

R-squared measures how well the regression model explains the variation in the dependent variable.

- Value ranges from 0 to 1
- Higher value means better model performance
- $R^2 = 1$ indicates perfect prediction
- $R^2 = 0$ indicates the model explains no variability

Question 10: How do you interpret the coefficients in a simple linear regression model?

In Simple Linear Regression:

- Intercept (a):

Represents the predicted value of Y when X = 0.

- Slope (b):

Represents the change in Y for every one-unit increase in X.

Example:
If the equation is:
$Y = 5 + 2X$

Then:
- 5 is the intercept
- 2 means Y increases by 2 units whenever X increases by 1 unit.

Question 9: Write Python code to fit a simple linear regression model using scikit-learn and print the slope and intercept.

(Include your Python code and output in the code box below.)

Answer:

Python Code:
```
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
Y = np.array([2, 4, 5, 4, 5])

model = LinearRegression()
model.fit(X, Y)
```

```
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
```

Output:
Slope: 0.6
Intercept: 2.2